RESQ-AI: Turning emergency information into faster action

GitHub

An AI-Assisted Disaster Response & Intelligence Platform

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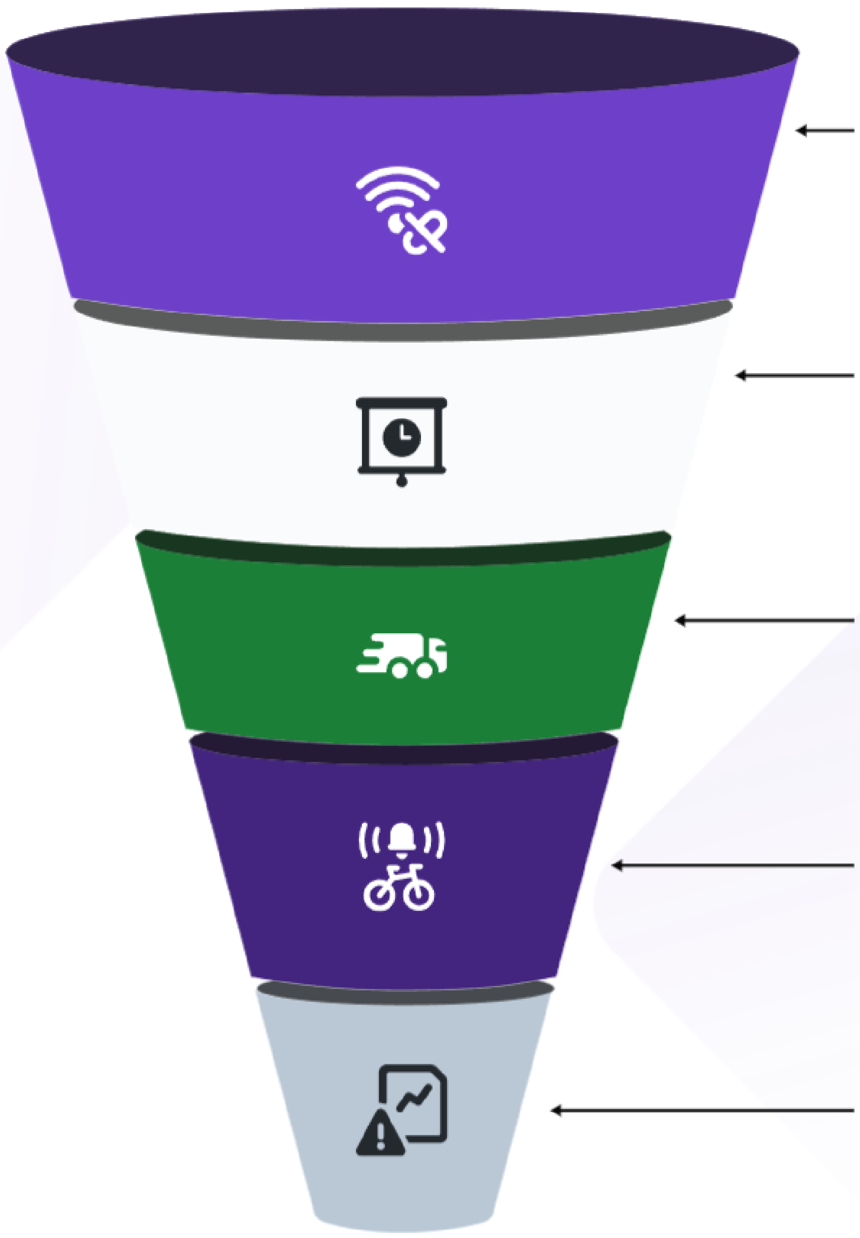
PROBLEM

Information is not enough: decisions must be fast

The critical gap in disaster response is turning scattered data into actionable intelligence.



Fragmented Sources

Emergency reports arrive via multiple channels, creating a "fog of war" for authorities.

Delayed Prioritization

Without automated triage, commanders spend critical minutes identifying incidents.

Resource Friction

Limited rescue resources are often misallocated due to a lack of centralized visibility.

Response Latency

Every second lost in communication increases the risk to affected lives and property.

The Core Problem

We don't lack data during a disaster; we lack the speed to act on it effectively.

RESQ-AI: Centralizing operational decisions

A unified platform connecting citizens, commanders, and volunteers in a single intelligence loop.

1

Incident Intelligence

Real-time monitoring of citizen-reported SOS signals with embedded location context.

2

Risk Prioritization

Automated assessment of emergency severity based on environmental and situational data.

3

Decision Support

AI-assisted analysis that helps commanders plan and assign rescue actions faster.

4

Resource Coordination

A role-based workflow that ensures the right volunteer reaches the right location instantly.

5

Unified Vision

One platform replaces the mess of phone calls, manual maps, and disconnected spreadsheets.

Our 6-step workflow for rapid response

A streamlined path from the initial SOS to a completed rescue operation.

Step	Action	Description
01	Report	Citizen submits emergency info via the RESQ-AI portal
02	Locate	Browser GPS and location-aware triggers add vital spatial context
03	Assess	The Risk Engine evaluates incident severity and weather context
04	Prioritize	Urgent situations are automatically flagged for the Commander
05	Coordinate	The Commander assigns specific actions or volunteers to the task

Empowering citizens with location-aware SOS tools

Providing victims with a direct line to help while gathering critical metadata automatically.

One-Tap SOS

Simplified reporting interface designed for high-stress emergency situations.

Automatic Context

Built-in GPS capture and Open-Meteo weather integration for immediate situational awareness.

Transparency

Citizen report tracking allows users to see the status of their request in real-time.

Guidance

Integrated emergency playbooks and shelter information to provide immediate self-help steps.

COMMANDER DASHBOARD

The Command Center: one view for operational mastery

Eliminating tool-switching by providing a single source of truth for decision-makers.

Real-time Monitoring

Live map visualization using Leaflet and OpenStreetMap for incident tracking.

Priority Triage

A dedicated feed highlighting incidents that require immediate commander attention.

Risk Visualization

Heatmapping and risk-zone identification based on AI-assisted analysis.

Resource Management

Direct interface to assign tasks to available volunteers based on proximity and skill.

Human-in-the-loop AI: decision support vs. automation

Our AI enhances human judgment rather than replacing the expertise of rescue commanders.

Safety First

Human oversight in all scenarios.

Decision

Commander makes final call.

Analysis

AI-assisted situation analysis.

Risk Engine

Python logic calculates priorities.

Input Layer

Correlates incident, location, data.

Closing the loop with volunteer coordination

Bridging the final mile of disaster response through role-based resource management.

Phase 1

Targeted Assignment

Commanders push specific incidents to volunteers' mobile-ready dashboards.

Phase 2

Acceptance Workflow

Volunteers can review incident details and location before accepting the task.

Phase 3

Status Tracking

Real-time updates (Accepted → In-Progress → Completed) sync back to the Command Center.

Phase 4

Role-Based Access

Secure authentication ensures only verified volunteers can access sensitive rescue data.

Phase 5

Accountability

Detailed metrics on response times and task completion help optimize future operations.

Technical architecture for reliability and speed

A modern stack combining React, FastAPI, and real-time intelligence engines.

Technical Stack

Frontend: React.js, TypeScript; Backend: FastAPI; Database: SQLite

Intelligence Engine

Python-based Risk Engine and AI decision-support logic

Citizen-to-Rescue

Centralized end-to-end workflow coordination

Location Intelligence

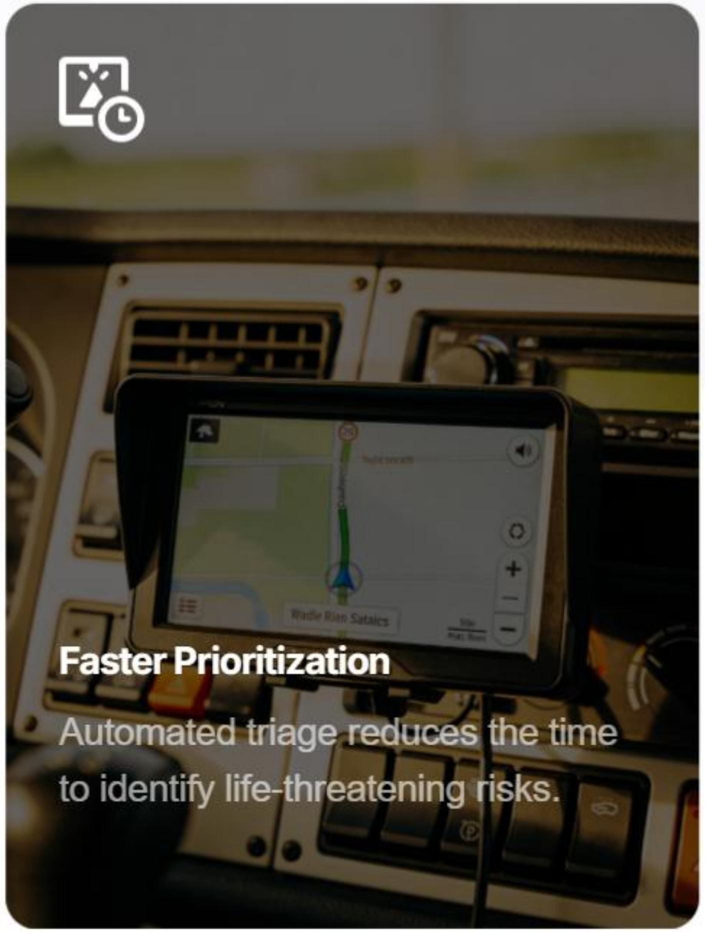
Leaflet, OSM, and Open-Meteo API integration

Human-in-the-loop

Specialized decision architecture for expert oversight

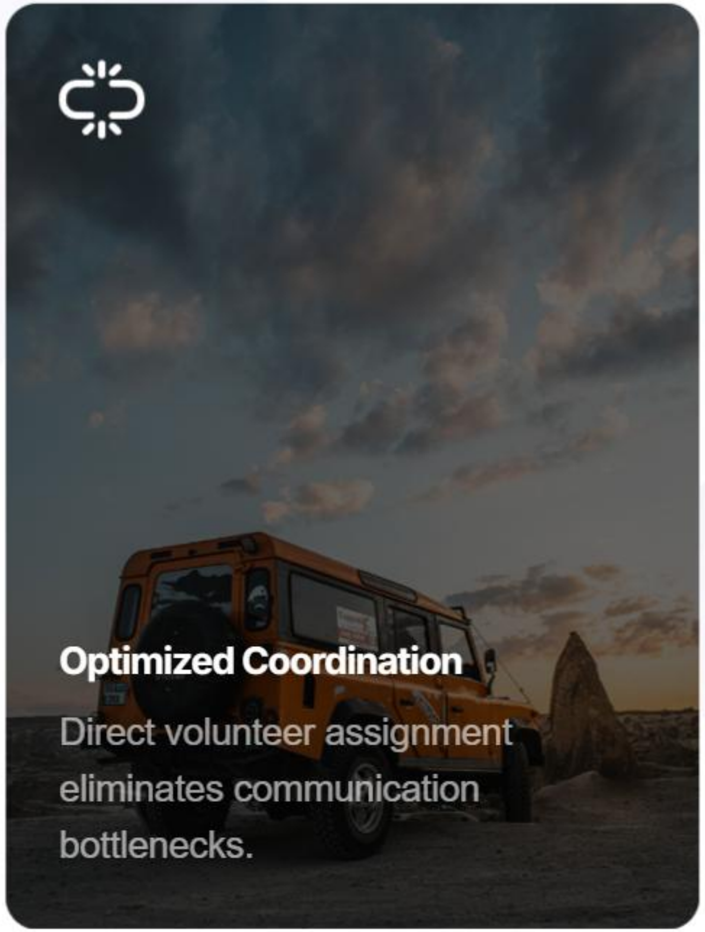
Less time finding: more acting

RESQ-AI helps the rescue team decide where to act first when every second counts.



Faster Prioritization

Automated triage reduces the time to identify life-threatening risks.



Optimized Coordination

Direct volunteer assignment eliminates communication bottlenecks.

01

Scenario: Flood Emergency

02

Flow: SOS to Response